



EEI6373 MINI PROJECT

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Evaluation of a Bank Customer Service System Using Queueing Theory

1. Introduction

Banks provide a wide range of financial services such as deposits, withdrawals, loan inquiries, and account management. During peak business hours, customers often experience long waiting times before receiving service. Excessive waiting not only reduces customer satisfaction but also affects employee productivity and the overall efficiency of the bank. Therefore, evaluating the performance of a bank customer service system is important for identifying bottlenecks and improving service quality.

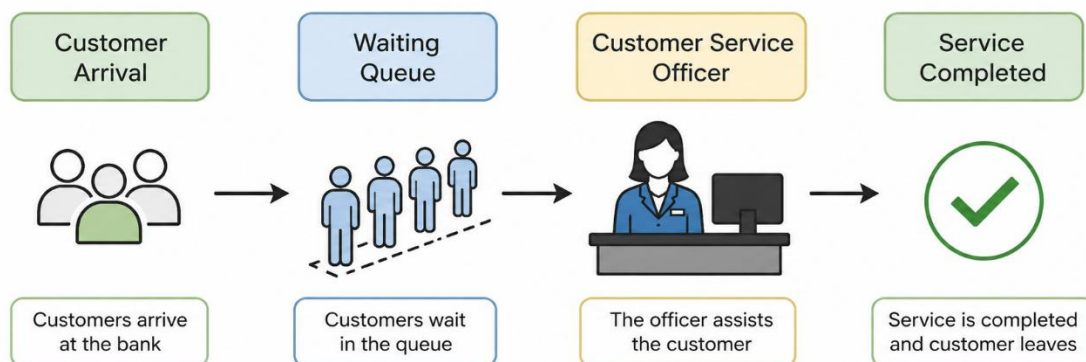
This report presents a performance evaluation of a bank customer service system, with a focus on minimizing customer waiting time using Queueing Theory. The study focuses on measuring customer waiting time, utilization, and throughput under different customer arrival rates. A simulated dataset is used to demonstrate the behavior of the system and to propose recommendations for improving performance.

2. System Description and Performance Goals

2.1 System Description

The selected system is the customer service counter of a commercial bank. Customers arrive randomly throughout the day and wait in a queue until a customer service representative becomes available. After completing the service, the customer leaves the system.

The process can be represented as follows



The system consists of

- One customer service officer
- A single waiting line
- First Come, First Served (FCFS)

2.2 Performance Goals

- Minimize customer waiting time.
- Reduce the average queue length.
- Maximize system throughput.
- Evaluate resource utilization.
- Identify potential bottlenecks during busy periods.

3. Modeling Approach and Assumptions

The M/M/1 queueing model was selected because it accurately represents a single server service system where customer arrivals are random.

The following assumptions are made

- Customer arrivals follow a Poisson distribution.
- Service times follow an exponential distribution.
- Only one customer service officer is available.
- Customers are served in the order they arrive.
- Customers do not leave the queue before receiving service.
- The queue has unlimited capacity.

These assumptions simplify the analysis while providing meaningful performance measurements.

In practice, bank service times are unlikely to follow a single exponential distribution, since transaction types vary widely a balance inquiry may take under a minute, while a loan application or account dispute can take five to ten minutes. This variability means the pure M/M/1 model, which assumes exponentially distributed service times, is a simplification. A more rigorous approach would use the M/G/1 queueing model, which allows service times

to follow any general distribution with a known mean and variance. Under M/G/1, the average queue length is given by the Pollaczek–Khinchine formula

$$L_q = [\lambda^2 \cdot E(S^2)] / [2(1 - \rho)]$$

where $E(S^2)$ is the second moment of the service time distribution. Because this formula explicitly accounts for the variance in service times, it captures the effect of long transactions inflating queue length in a way that M/M/1, which only uses the mean service rate, cannot. For this study, M/M/1 was retained for analytical simplicity and because it provides a reasonable first-order approximation, but M/G/1 would offer a more realistic model of teller behavior in future work.

4. Data Description and Methodology

Since actual banking data was unavailable, based on simulated data realistic customer arrival patterns observed during banking hours. Arrivals were recorded in 30-minute intervals across the full working day (9:00 to 17:00) to obtain a reliable estimate of the average arrival rate.

time_period	customers_arrived
9:00-9:30	18
9:30-10:00	22
10:00-10:30	26
10:30-11:00	31
11:00-11:30	24
11:30-12:00	19
12:00-12:30	35
12:30-13:00	32
13:00-13:30	27
13:30-14:00	18
14:00-14:30	13
14:30-15:00	14
15:00-15:30	24
15:30-16:00	34
16:00-16:30	25
16:30-17:00	22
Total customers	384

Average arrival rate

$$\lambda = 48 \text{ customers per hour}$$

The customer service officer serves approximately

$$\mu = 60 \text{ customers per hour}$$

The service rate was assumed to be 60 customers per hour, based on an average service time of approximately one minute per customer. This value is used as a simplified assumption for the M/M/1 model to demonstrate the performance of the customer service system.

5. Queueing Model Calculations

Server Utilization

The utilization factor is calculated using

$$\rho = \lambda / \mu$$

$$\rho = 48 / 60$$

$$\rho = 0.80$$

This indicates that the customer service officer is busy approximately 80% of the working time.

Average Number of Customers in the Queue

$$L_q = \lambda^2 / [\mu(\mu - \lambda)]$$

$$L_q = 48^2 / [60(60 - 48)]$$

$$L_q = 2304 / 720$$

$$L_q = 3.2 \text{ customers}$$

Average Number of Customers in the System

$$L = \lambda / (\mu - \lambda)$$

$$L = 48 / 12$$

L = 4 customers

Average Waiting Time in Queue

$$W_q = \lambda / [\mu(\mu - \lambda)]$$

$$W_q = 48 / 720$$

$$W_q = 0.0667 \text{ hours}$$

$$= 4 \text{ minutes}$$

Average Time Spent in the System

$$W = 1 / (\mu - \lambda)$$

$$W = 1 / 12$$

$$= 0.0833 \text{ hours}$$

$$\approx 5 \text{ minutes}$$

6. Performance Improvement Recommendations

The queueing analysis indicates that the bank's customer service system operates at 80% utilization, with an average waiting time of approximately 4 minutes and an average of 3.2 customers waiting in the queue. Although the current system is capable of handling the average arrival rate of 48 customers per hour, the high utilization means that the system may experience significant congestion when customer arrivals increase during peak periods. Based on these findings, several improvements can be considered.

➤ Employ an Additional Customer Service Officer During Peak Hours

One of the most direct ways to reduce customer waiting time is to increase the number of available service officers during periods of high demand. The simulated data shows that customer arrivals are not constant throughout the day. For example, 35 customers arrived between 12:00 and 12:30, while 34 customers arrived between 15:30 and 16:00. These periods indicate higher customer demand compared with some other time intervals.

Instead of keeping the same number of officers throughout the entire working day, the bank could assign an additional officer during these peak periods. Increasing service capacity allows more customers to be served simultaneously and reduces the number of customers waiting in the queue.

For example, if one officer can serve approximately 60 customers per hour, adding a second officer would significantly increase the available service capacity. This approach would be particularly useful during periods where the arrival rate temporarily approaches or exceeds the capacity of the existing service counter.

However, employing an additional officer for the entire working day may not be cost-effective because customer arrivals are lower during some periods. Therefore, a flexible staffing strategy in which additional employees are scheduled only during peak periods would provide a better balance between service quality and operational cost.

➤ Introduce an Appointment Scheduling System for Complex Services

Not all banking transactions require the same amount of service time. Simple activities such as balance inquiries or basic deposits may be completed quickly, whereas services such as loan applications, account disputes, and financial consultations may require significantly more time.

Long transactions can delay other customers because the service officer remains occupied for an extended period. This can increase the waiting time of customers who are waiting behind them.

An appointment scheduling system could be introduced for complex services. Customers requiring loan consultations, account-related investigations, or other time-consuming services could book a specific time in advance. This would reduce the number of complex transactions entering the normal walk-in queue.

The appointment system would also allow the bank to predict the expected workload and allocate employees accordingly. As a result, the normal queue could be reserved mainly for shorter transactions, helping to reduce average waiting time.

➤ Encourage Customers to Use Online and Mobile Banking Services

Another way to reduce pressure on physical bank branches is to encourage customers to use digital banking services. Many simple transactions do not necessarily require customers to visit a branch.

Examples include:

- Checking account balances
- Viewing transaction history

- Transferring money
- Paying bills
- Managing certain account details
- Checking transaction information

By moving these simple transactions to online or mobile banking platforms, the number of customers physically arriving at the branch can be reduced. A lower arrival rate would directly reduce the workload placed on the service counters.

For example, the current average arrival rate is 48 customers per hour. If digital banking reduces physical branch arrivals, the effective arrival rate could decrease, resulting in lower server utilization and shorter waiting times.

Therefore, digital banking should not only be viewed as a convenience for customers but also as a method of improving the performance of the physical branch service system.

➤ Monitor Customer Arrival Patterns to Optimize Staff Scheduling

The simulated dataset demonstrates that customer arrivals vary throughout the working day. For example, the number of arrivals increases during certain periods and decreases during others. Therefore, assigning the same number of employees throughout the entire day may not be the most efficient staffing strategy.

The bank could continuously collect customer arrival data and identify periods with consistently high demand. Staff schedules could then be adjusted according to these patterns.

For example, if the bank identifies that customer arrivals are consistently high around lunchtime and late afternoon, additional employees could be scheduled during these periods. During low-demand periods, fewer employees may be required.

This approach would improve resource utilization while reducing waiting times. It also prevents the bank from unnecessarily increasing staffing costs during periods when customer demand is low.

The queueing analysis suggests that the current system is operating at approximately 80% utilization. Although this is manageable under average conditions, the system has limited spare capacity when customer arrivals increase. Therefore, the bank should not rely on a single improvement.

A combination of strategies would provide better results. Additional staff could be introduced during peak periods, while appointment scheduling, digital banking, and self-

service kiosks could reduce the number of customers entering the physical service queue. At the same time, continuous monitoring of arrival patterns would allow management to adjust staffing levels according to actual demand.

These improvements could reduce customer waiting time, prevent excessive queue formation, improve employee utilization, and increase overall customer satisfaction.

7. Visualizations and Discussion

Figure 1 presents the number of customers arriving at the bank during each 30 minute interval from 9:00 to 17:00. The graph shows that customer arrivals vary throughout the working day rather than remaining constant. The highest number of arrivals occurs between 12:00 and 12:30, with 35 customers, followed by 34 customers between 15:30 and 16:00. The lowest number of arrivals occurs between 14:00 and 14:30, with 13 customers.

This variation indicates that customer demand is higher during certain periods of the day. Therefore, using the same staffing level throughout the entire working day may not be the most efficient approach. Additional staff could be allocated during high arrival periods to reduce queue formation and customer waiting time.

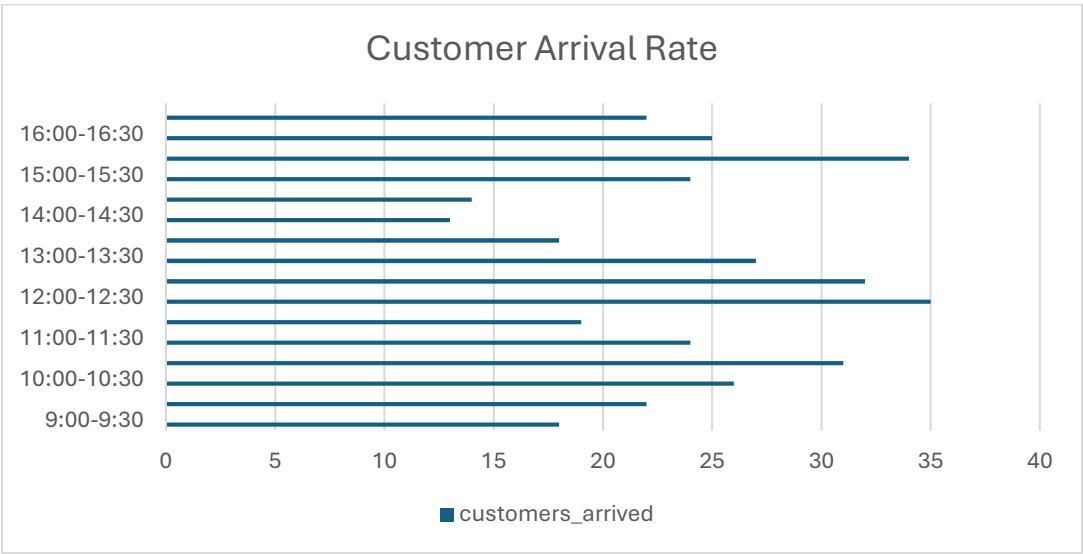


Figure 2: Waiting Time versus Arrival Rate

Figure 2 illustrates the relationship between the customer arrival rate and the average waiting time in the queue. The graph demonstrates that waiting time increases as the arrival rate approaches the service capacity of the system.

At lower arrival rates, the service officer has sufficient capacity to handle incoming customers, resulting in shorter waiting times. However, as the arrival rate increases, the service officer becomes more heavily utilized and customers spend more time waiting in the queue. When the arrival rate gets close to the service rate, the waiting time increases significantly.

For the current system, the average arrival rate is 48 customers per hour and the service rate is 60 customers per hour. This results in a utilization of 80%. Therefore, although the system can handle the average demand, a temporary increase in customer arrivals during peak periods can cause congestion and increase waiting times.

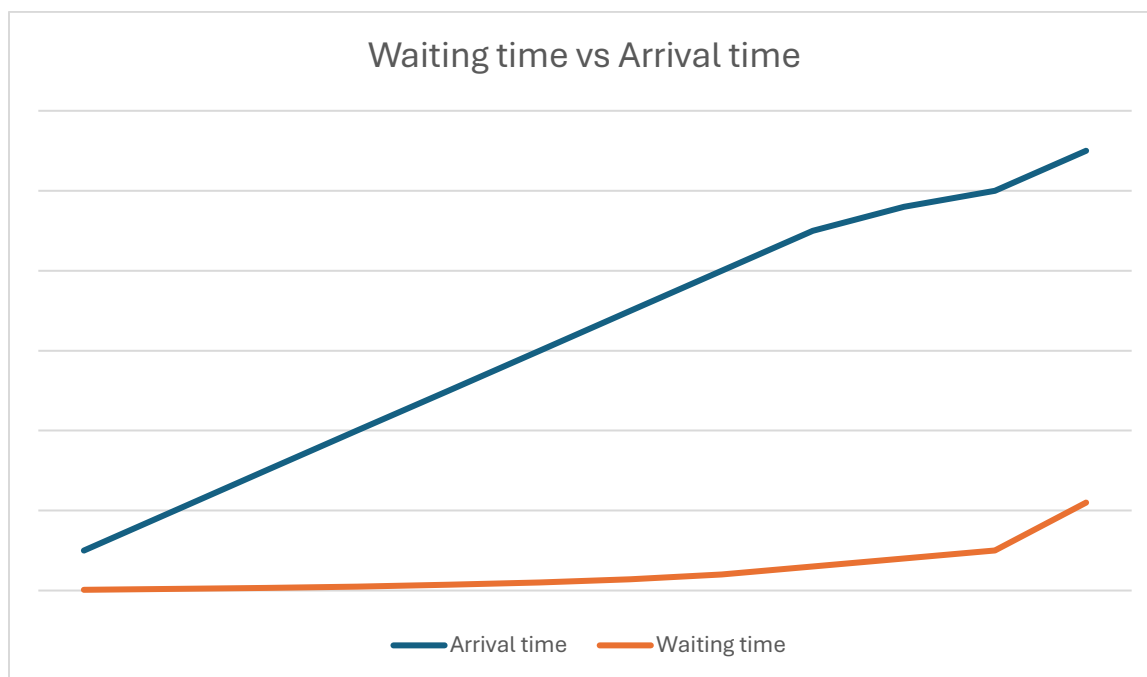
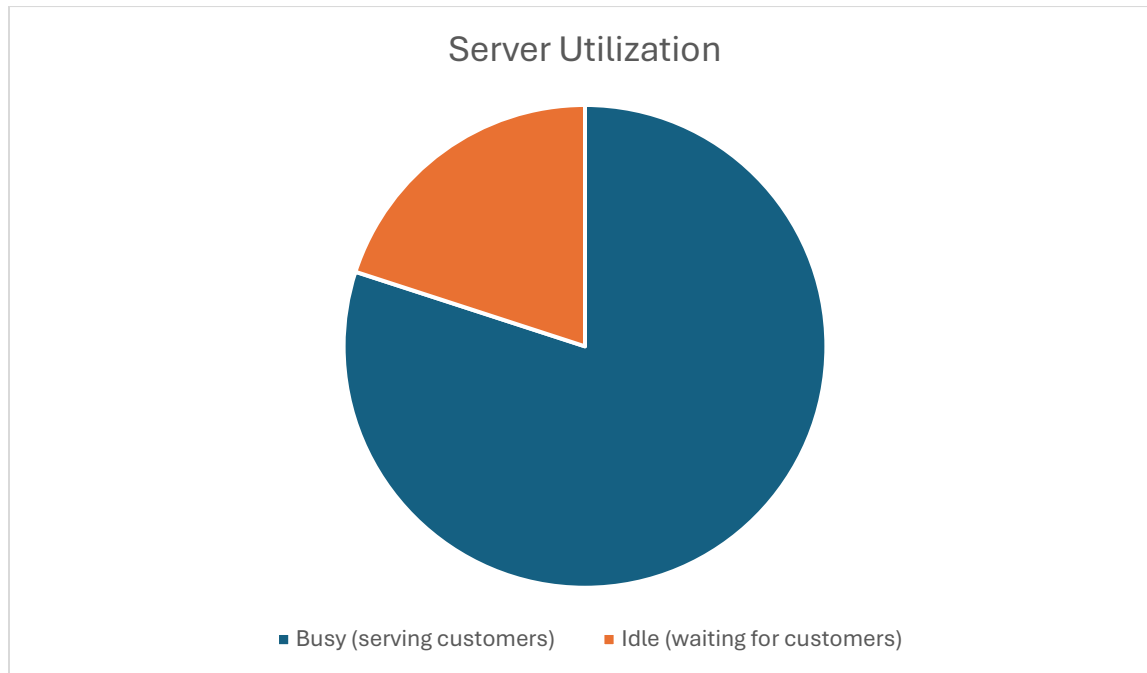


Figure 3: Server Utilization

Figure 3 presents the utilization of the customer service officer. The calculated utilization is 80%, meaning that the service officer is expected to be busy approximately 80% of the time and idle for the remaining 20%.

An 80% utilization indicates that the available service capacity is being used relatively efficiently. However, it also means that the system has only limited spare capacity to handle sudden increases in customer arrivals. During peak periods, when the arrival rate temporarily increases, the queue may grow because customers arrive faster than they can be served.

Therefore, the utilization result supports the recommendation of adjusting staffing levels during peak periods. Adding temporary service capacity when demand is high could reduce waiting times without requiring additional staff throughout the entire working day.



8. Limitations

- ❖ **Simulated data,** The analysis uses simulated customer arrival data rather than actual records from a bank. Although the data was designed to represent realistic customer arrival patterns, it may not fully represent the behavior of a real bank branch.
- ❖ **Single-server assumption,** The M/M/1 model assumes that one customer service officer handles the queue. In a real bank, multiple service officers may operate simultaneously, particularly during peak periods.
- ❖ **Simplified service rate,** A constant service rate of 60 customers per hour was assumed for the analysis. In reality, service times can vary depending on the type and complexity of the banking transaction.
- ❖ **Exponential service-time assumption,** The M/M/1 model assumes that service times follow an exponential distribution. However, transactions such as loan inquiries, account disputes, and account opening may require considerably more time than simple transactions.

- ❖ **Customer behavior**, The model does not consider customers leaving the queue because of long waiting times, joining another queue, or using alternative banking channels.

9. Future Extensions

- ❖ **Multiple service counters**, Future analysis could use an M/M/c model to represent multiple customer service officers working simultaneously. This would allow the effect of adding additional officers during peak hours to be evaluated quantitatively.
- ❖ **Real banking data**, Actual customer arrival and service-time data could be collected to obtain more accurate estimates of arrival rates, service rates, waiting times, and queue lengths.
- ❖ **Variable service times**, An M/G/1 or M/G/c model could be considered because different banking transactions require different amounts of service time. This would provide a more realistic representation of transaction variability.
- ❖ **Simulation based analysis**, Simulation tools such as SimPy, Arena, or AnyLogic could be used to model changing arrival rates, different staffing levels, and peak-hour conditions.
- ❖ **Peak hour staffing analysis**, Future work could compare different staffing configurations, such as one, two, or three service officers, to determine the most effective balance between waiting time and staffing costs.

10. Conclusion

This study evaluated the performance of a bank customer service system using an M/M/1 queueing model. Based on the simulated data, 384 customers arrived during the eight-hour observation period, resulting in an average arrival rate of 48 customers per hour. With an assumed service rate of 60 customers per hour, the system utilization was calculated as 80%. The model estimated an average queue length of 3.2 customers and an average waiting time of approximately 4 minutes.

The results indicate that the system can handle the average customer demand under normal conditions. However, the relatively high utilization means that the queue can grow during periods of increased demand. The dataset also demonstrates variations in customer arrivals throughout the working day, highlighting the importance of managing peak-hour demand.

Based on these findings, strategies such as adding staff during peak periods, introducing appointment scheduling for complex services, promoting digital banking, and providing self

service facilities could help reduce waiting times and improve resource utilization. Further analysis using multiple server and general-service-time queueing models would provide a more realistic representation of the bank's operations.

11. References

[1] Open University of Sri Lanka, EEI6373 Performance Modelling - Mini Project Instructions, 2026.